

Extension: Contextual Bandits

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Extends the MVP bandits (§5 of `mvp_specification.tex`) by conditioning action selection on the current activity level.

1 Formulation

Context $c_t = s_t.\text{activity} \in \{\text{sedentary}, \text{active}\}$. Each algorithm maintains separate parameters per (c, a) pair instead of global per-action parameters:

- **Thompson Sampling:** $(\alpha_{c,a}, \beta_{c,a})$ posteriors. Sample $\theta_{c_t,a} \sim \text{Beta}(\alpha_{c_t,a}, \beta_{c_t,a})$, pick $a^* = \arg \max_a \theta_{c_t,a}$.
- **Epsilon-Greedy:** $Q_{c,a}$ values. With probability ϵ explore, otherwise pick $a^* = \arg \max_a Q_{c_t,a}$.
- **UCB:** $Q_{c,a}$ values with per-context counts. $a^* = \arg \max_a [Q_{c_t,a} + c\sqrt{\ln(N)/n_{c_t,a}}]$ where N is global step count.

Update only the observed (c_t, a_t) pair.

2 Optimal Policy

The optimal contextual policy (from the transition matrices in §3) is: nudge when sedentary ($0.3 > 0.1$), idle when active ($0.6 > 0.5$). Stationary distribution: $\pi_s = 4/7$, $\pi_a = 3/7$.

$$E[r] = \frac{4}{7} \cdot 0.3 + \frac{3}{7} \cdot 0.6 = \frac{3}{7} \approx 0.429 \text{ per step} \implies 192.9 \text{ total}$$

Best non-contextual (always nudge): $E[r] = 0.375$, total 168.8. The contextual bound is 14.3% higher.

3 Results

50 seeds, 450 timesteps. Full methodology follows §6 of `mvp_specification.tex`.

Per-step reward in the last 50 steps: contextual TS reaches 0.429 (0.0% from optimal), contextual EG reaches 0.414 (3.4% from optimal), contextual UCB reaches 0.366 (14.4% from optimal). All contextual agents outperform their standard counterparts. Thompson Sampling benefits most from context, followed by epsilon-greedy; UCB's per-context exploration bonus improves over the global baseline but the small action space limits the gain.

Agent	Total Reward	Per Step	Last 50 Steps	vs Contextual Opt
Contextual TS	178.7 ± 15.8	0.397	0.429	-7.4%
Standard TS	159.6 ± 12.7	0.355	0.373	-17.2%
Contextual EG	171.6 ± 20.1	0.381	0.414	-11.0%
Standard EG	158.1 ± 17.8	0.351	0.370	-18.0%
Contextual UCB	159.7 ± 14.1	0.355	0.366	-17.2%
Standard UCB	148.4 ± 12.6	0.330	0.354	-23.1%
Contextual optimal	192.9	0.429	0.429	—
Non-contextual optimal	168.8	0.375	0.375	—

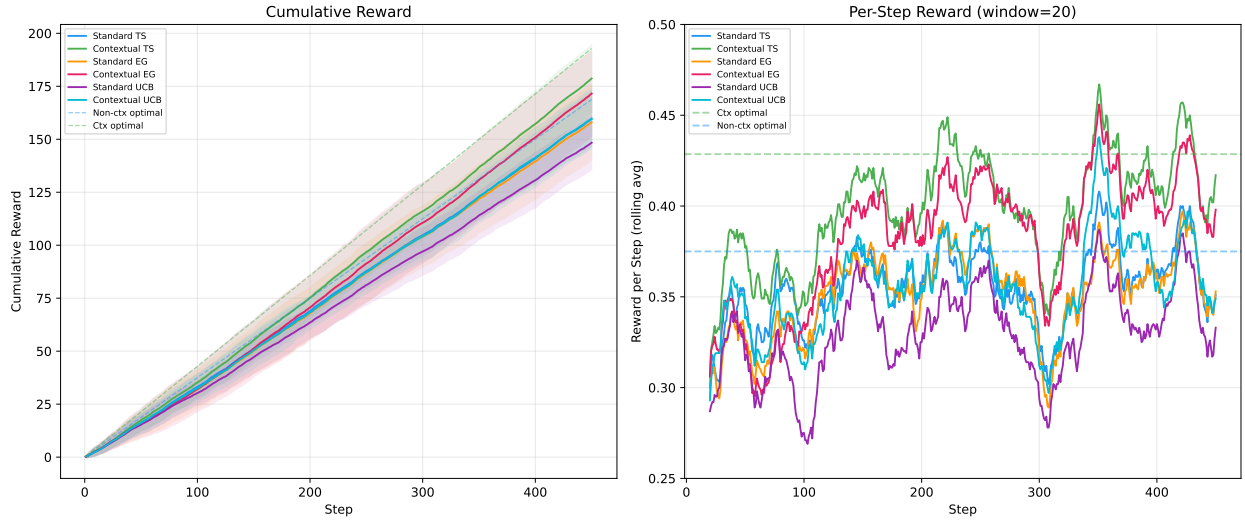


Figure 1: Left: cumulative reward (50 seeds, ± 1 SD). Right: per-step reward (20-step rolling avg). Contextual agents converge toward $3/7 \approx 0.429$; standard agents plateau at lower rates.